Optics Communications 473 (2020) 125699

Contents lists available at ScienceDirect

Optics Communications

journal homepage: www.elsevier.com/locate/optcom

A hybrid VLP and VLC system using *m*-CAP modulation and fingerprinting algorithm

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ARTICLE INFO

Keywords:
Visible light positioning (VLP)
Hybrid VLP and VLC
Indoor positioning system (IPS)
m-CAP modulation
Fingerprinting

ABSTRACT

A global positioning system (GPS) with high positioning accuracy (PA) is not feasible for challenging environments such as tunnels, underground, complex buildings, etc. For such environments, the alternative technology of visible light positioning (VLP) system has emerged as a viable solution. In this paper, we propose for the first time, a hybrid multi-band carrier-less amplitude and phase (*m*-CAP) based VLP and visible light communications (VLC) system using the fingerprinting algorithm for determining the position of the receiver, i.e., VLP, and employing the unused sub-bands of *m*-CAP for VLC. We compare the results for different transmitter configurations and a range of step sizes in terms of the position error (PE) and the bit error rate (BER) performances. Results show that, PE increases with the step size, which is the most critical parameter in fingerprinting-based positioning systems, for a fixed bit energy to noise ratio (E_b/N_0). For a fixed E_b/N_0 of 10 dB the PE values are 4.5 and 10.7 cm for the step sizes of 1 and 20 cm, respectively. For a fixed step size, in contrast, PE decreases with increasing E_b/N_0. Numerical results for a fixed step size of 5 cm show that the PE is reduced from 13.8 to 3.3 cm by 10 dB increase in E_b/N_0. In addition, by increasing the step size, the quantization error overcomes the noise-induced error. We also show that, configurations I and II outperform configuration III in terms of the PE by about 8 to 6 cm at E_b/N_0 of 5 dB and BER (i.e., $1.56{\times}10^{-4}$, $2.52{\times}10^{-4}$ and $2.54{\times}10^{-3}$ for configurations I, II and III, respectively at E_b/N_0 of 20 dB), since configuration III is approximately linear with a symmetrical fingerprint map and a lower illumination level.

1. Introduction

The 5th generation (5G) and beyond wireless communications will support a diverse range of methods and techniques for different applications and scenarios delivering high quality services [1]. Despite its low security at the physical layer and relatively low data rates, the radio frequency (RF) wireless technologies are still the most widely used for communications due to its wide coverage area and high mobility. However, with the increasing growth of wireless network traffics and the need for more bandwidth, the RF technology is experiencing spectral congestions, which needs to be addressed. Hence, finding and deploying the new spectrum bands can be effective to prevail with this problem [2,3]. There are a number of possible options to address this problem including shifting the transmission carrier frequency to the millimeter (mm)-wave and/or optical frequency band. The latter is attractive since it offers a very high bandwidth, inherent security, lack of susceptibility to the RF electromagnetic interference, and low power usage compared with the RF technologies [3–8]. Optical wireless communications (OWC) covering all three optical bands of ultra violet (UV), visible, and infrared (IR), is an alternative complementary technology to RF that could be used in multitude of applications in both indoor and outdoor environments [9]. The OWC technology employing the visible band is known as visible light communications (VLC), which uses solid-state (SS) lights as an optical transmitter (Tx) (i.e., optical antenna) to provide simultaneous data communications, illumination and positioning services [3,5].

The use of VLC is significantly increasing for a number of reasons including high-level security in the physical layer since the light does not pass through walls and other opaque obstacles, high throughput and low latency in urban environments with high density traffic due to the use of pico- and femto-cells, low power base-stations (BSs) and high data rates using massive multiple-input multiple-output (MIMO) [1,3]. Moreover, with the growing interest in smart living, all possible light emitting diodes (LEDs)-based lights in indoor and outdoor environments, traffic lights, etc., could be used for data communications, sensing and positioning [1]. The latter i.e., the visible light positioning (VLP) system is one of the promising applications, which has received the interest of researchers in recent years [4–8,10,11]. For outdoor environments, the global positioning system (GPS) offers a good coverage

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https://doi.org/10.1016/j.optcom.2020.125699
Received 25 December 2019; Received in revised form 5 March 2020; Accepted 6 March 2020
Available online 7 March 2020

with sufficient accuracy (i.e., >10 m) and low cost [10]. However, GPS cannot be used in indoor environments such as tunnels, undergrounds, and shopping centers, etc., due to the received weak signals or no signals from satellites [4,11].

Indoor positioning systems (IPSs) require more accuracy compared with the outdoor environments. In recent years, in order to replace GPS in indoor environment and improve its positioning accuracy, a range of IPSs have been proposed based on different technologies including ultrasound, wireless local area networks (WLAN), ultra-wideband (UWB), WiFi [12], Bluetooth [13], radio frequency identification (RFID) [14], and ZigBee [15]. The RF-based systems suffer from multipath induced fading due to reflections from walls and objects within rooms, and offer limited positioning accuracy (PA), which may not be desirable in certain indoor applications [16–20]. The operating range of RFID and UWB are limited [14] and Bluetooth requires user association, which is an undesired feature [13]. However, since WiFi and Bluetooth are widely used in smart devices they are often used for IPS [4].

The VLP systems use existing lighting fixtures, i.e., LEDs, with some modifications to make them act as optical antennas or access points for transmitting data information and positioning coordinates. LED-based VLP systems are very accurate with a positioning error (PE) of <1 cm [4] compared with 1–7 and 2–5 m in WiFi and Bluetooth-based systems, respectively. Lighting systems have inherent spatial distribution, with the position of the light sources being optimized to comply with lighting uniformity constraints [21]. In addition, the propagation of light beam is more predictable due to the reduced impact of multipath induced fading than RF-based links [7,8]. The security of VLP is similar to that of VLC, where it can be used in locations such as hospitals without affecting the sensitive medical equipment. VLP systems are perfect for environments such as underwater, underground mines, tunnels, etc. [11]. In addition, despite the advantages of LED-based VLP, there are a number of drawbacks including the requirement for the line of sight (LOS) paths and locations of the LED lights [22].

There are many IPSs for VLP reported in the literature including the angle of arrival (AOA), time of arrival (TOA), time difference of arrival (TDOA), phase difference of arrival (PDOA) and received signal strength (RSS) [21]. The AOA system, which uses the receiver (Rx) diversity to compute the angle of arrival information [7,23], offers reduced PE but at the cost of increased computational time; while both TOA and TDOA require that signals are synchronized [24]; however PDOA offers a lower computational complexity than TDOA but needs a local oscillator (LO). RSS has relatively low PA because of the ambient light induced noise and the tilt angle of the photodiode (PD), thus requiring a higher signal to noise ratio (SNR) [25]. Note, since RSS values can easily be extracted using a PD without the need for a secondary device, therefore it has been extensively used in VLP and is also adopted in this work. However, the received signal intensity is inversely proportional to the square of distance, d^2, between the Tx and the Rx. Accordingly, various algorithms including trilateration, fingerprinting and proximity, which are described in next section, have been developed for VLP [4].

Authors in [10] proposed a two-dimensional (2D) visible light-based IPS using fingerprinting of received optical signals emitted from LEDs. In this scheme, each Tx was assigned with a different frequency and the power spectral density (PSD) of the received signal was calculated at the Rx and the position was estimated (to within a few centimeters) as one of the fingerprint locations using a novel but simple algorithm. In [25], a novel indoor VLP model based on k nearest neighbors (kNN) was used as an efficient solution to expand the number of input features for the RF signal in a reflective environment. In [26], the development of a practical VLP system using RSS was reported, where it was shown that IPS is accurate, and easy to train and calibrate despite using the fingerprinting technique. Also investigated was the impact of distance metrics used to compute the weights of the weighted kNN (WkNN) algorithm on the PA at the cost of increased volume of offline measurements for fingerprint mapping.

In many indoor environments, a large number of LEDs are used for illumination. In such cases, the illumination areas overlap and therefore can lead to inter-cell interference (ICI) in VLP and VLC. To overcome ICI, frequency division multiplexing (FDM), time division multiplexing (TDM) and wavelength division multiplexing (WDM) schemes are adopted [27,28]. Despite the advantages of FDM and TDM, both protocols have their limitations. For example, in order to increase the spectral efficiency (SE) in FDM and introduce multiple-access (MA) capability, orthogonal FDM (OFDM) has been introduced in visible light-based IPSs, where the input data streams are encoded with assigned orthogonal subcarriers prior to intensity modulation (IM) of LEDs. In OFDM, a frequency-selective channel can be converted into a set of flat sub-bands, thus more resilient to multipath fading (in particular deep fading) in wireless channels [4,28]. In [28], a minimum of three subcarriers and therefore three LEDs were used for IPS with no ICI and with a mean pointing error (PE) of 1.3 cm. Nevertheless, OFDM has specific drawbacks including higher peak-to-average power ratio (PAPR), thus the need for highly linear amplifiers and LEDs, and complexity due to fast Fourier transform (FFT) and inverse FFT (IFFT). Alternatively, carrier-less amplitude and phase (CAP) modulation, which is simpler than OFDM and with lower PAPR as well as no FFT and IFFT could be adopted in VLC and VLP, where carrier frequencies are generated by carefully designed finite impulse response (FIR) filters. These filters can be produced in analog or digital formats, but the latter is preferred since they are low cost, easily reconfigurable, integratable and programmable [29]. However, CAP requires a flat frequency response, which is not always the case in VLC mainly due to the light source characteristics. To increase the systems' performance with a non-flat frequency response, a complex equalizer can be used. Note, splitting the available signal bandwidth into m sub-bands results in multi-band CAP (m-CAP), which can be used to overcome the flat band requirement and to reduce the PAPR compared with OFDM [30–32].

To the best of the authors' knowledge, m-CAP modulation has not been adopted in VLP systems. Therefore, in this paper we propose a fingerprinting-based VLP using m-CAP modulation. Furthermore, the unused sub-bands of m-CAP can be used for data communications, i.e., a hybrid VLP-VLC. We use three LEDs and a single PD as the Txs and the Rx, respectively. We show that, the Tx's configuration on the ceiling has an effect on the PE and the system bit error rate (BER) performances.

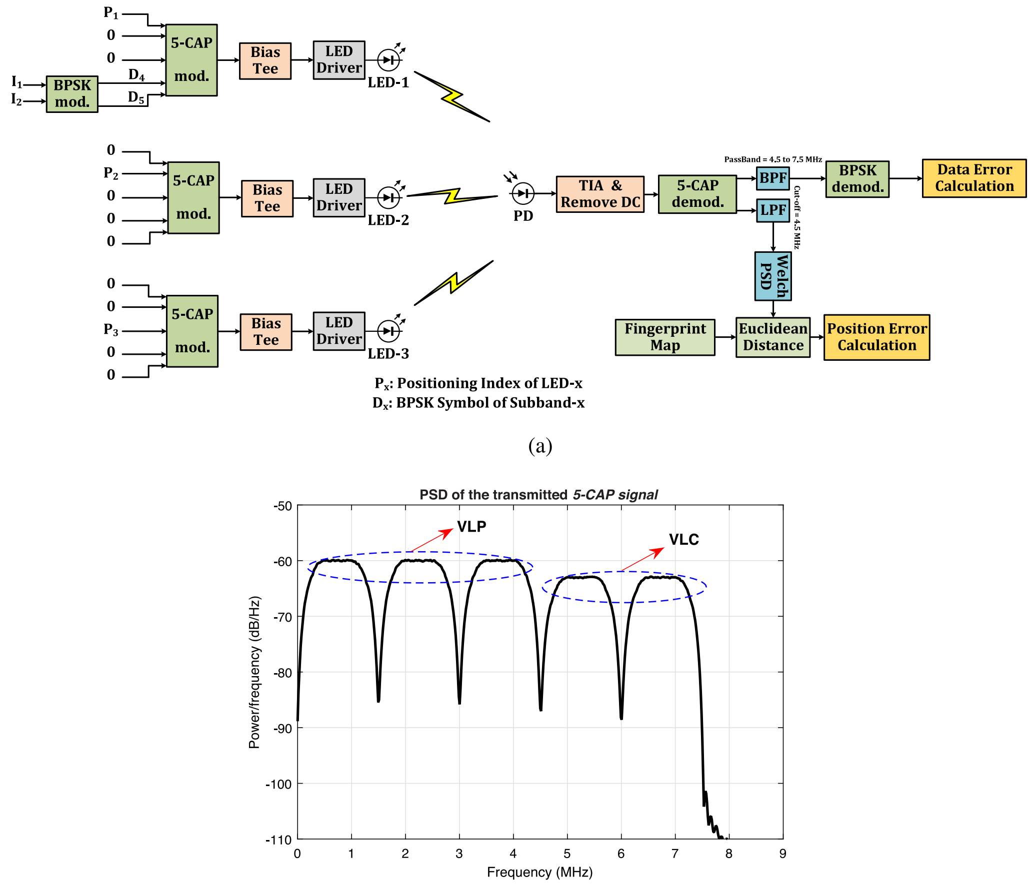
The rest of the paper is organized as follows. In Section 2 overview of the fingerprinting visible light-based IPS using m-CAP modulation is comprehensively explained. In Section 3, the most significant results are presented and discussed. Eventually, Section 4 is assigned to conclude the paper.

2. System overview

Fig. 1 shows the schematic system block diagram of the proposed system. In the following we describe each subsystem.

2.1. Transmitter

The performance of VLC and the visible light-based IPS are investigated simultaneously in this paper using m-CAP modulation. The main idea is to assign each sub-band of m-CAP to a Tx (an LED) for easy recognition at the Rx in indoor VLP. At least three LEDs should be used at the Tx in order to achieve accurate positioning, thus the use of 3-CAP. However, as shown in Fig. 1, we have used 5-CAP in the proposed hybrid VLP-VLC, where 3 and 2 sub-bands (i.e., P_1, P_2 and P_3; and D_4 and D_5) are assigned for VLP and VLC (i.e., data transmission), respectively. In the 1st CAP module, P_1 (i.e., the positioning index of LED$_1$) is inserted to the 1st sub-band and the 2nd and 3rd sub-bands are not utilized in order to avoid interference. The 4th and 5th sub-bands are allocated for data transmission, where the binary phase shift keying (BPSK) modulation format is adopted. P_2 and P_3 are inserted in the 2nd



(a)

(b)

Fig. 1. Hybrid VLP-VLC systems: (a) schematic diagram and (b) 5-CAP bands.

and 3rd sub-bands of CAP modules 2 and 3, respectively. It is also worth mentioning that larger m can be employed for this system. For higher order of m-CAP more sub-bands will be assigned to data transmission, thus increased data rates.

Note, symbols in each sub-band of m-CAP are first up-sampled by inserting $(n_s - 1)$ zeros between two successive symbols. More details on m-CAP can be found in [30] and the references within. The up-sampling factor is given as [33]:

$$n_s = \lceil 10(1+\beta) \rceil , \quad (1)$$

where $\lceil\ \rceil$ is the ceiling function and β is the roll-off factor of square root raised cosine (SRRC) filter used in CAP. The up-sampled symbols are then split into the real and imaginary parts, which are applied to the in-phase (I) and quadrature (Q) finite impulse response (FIR) filters, respectively, which are given by, respectively [30,33]:

$$f_I^n(t) = \left[\sin\left[\gamma(1-\beta)\right] + 4\beta(t/T_s)\cos\left[\gamma\delta\right] / \gamma \left[1 - \left(4\beta t/T_s\right)^2 \right] \right] . \cos\left[\gamma(2n-1)\delta\right] , \quad (2)$$

$$f_Q^n(t) = \left[\sin\left[\gamma(1-\beta)\right] + 4\beta(t/T_s)\cos\left[\gamma\delta\right] / \gamma \left[1 - \left(4\beta t/T_s\right)^2 \right] \right] . \sin\left[\gamma(2n-1)\delta\right] , \quad (3)$$

where T_s is the symbol duration, $\gamma = \pi t/T_s$ and $\delta = 1+\beta$. Note, in m-CAP carrier frequencies are generated by carefully designed FIR filters in analog or digital domains. Digital FIR filters are preferred because of low cost, easy re-configurability, integration, and programmability. Finally, all sub-bands are summed together to form the 5-CAP signal, which is given as [33]:

$$s(t) = \sum_{n=1}^{5} \left(s_I^n(t) \otimes f_I^n(t) - s_Q^n(t) \otimes f_Q^n(t) \right) , \quad (4)$$

where $s_I^n(t)$ and $s_Q^n(t)$ are the up-sampled in-phase and quadrature components of the n^{th} branch, respectively and $\otimes$ denotes the time domain convolution. The outputs of m-CAP modulators are used for intensity modulation (IM) of the LEDs via the bias-tee and drivers modules. Here, at least three LEDs (Txs) are required for positioning, and each LED is assigned with a sub-band of m-CAP. The impulse response of LED, which is modeled as a 1st order low-pass filter (LPF) with a 3-dB cut-off frequency of f_{LED}, is given as [32]:

$$h_{LED}(t) = e^{-2\pi f_{LED} t} . \quad (5)$$

As mentioned before, three LEDs are used in our proposed system for illumination, positioning and communications. As will be illustrated in Section 3, LEDs locations on the ceiling, i.e., LEDs configuration, will affect the performance of the system. In this work, we consider three types of LED configurations as shown in Fig. 2, and compare them in terms of the PE and BER, see Section 3.

2.2. Channel

The VLC channel is normally composed of LOS and diffuse or non-LOS (NLOS) components. Here, we only consider the LOS with a channel transfer function, which is given by [33]:

$$h_j = \begin{cases} \dfrac{(L+1)A}{2\pi d^2} \cos^L(\phi) T_s(\psi) \cos(\psi) , & 0 \le \psi \le \psi_c \\ 0, & \psi > \psi_c , \end{cases} \quad (6)$$

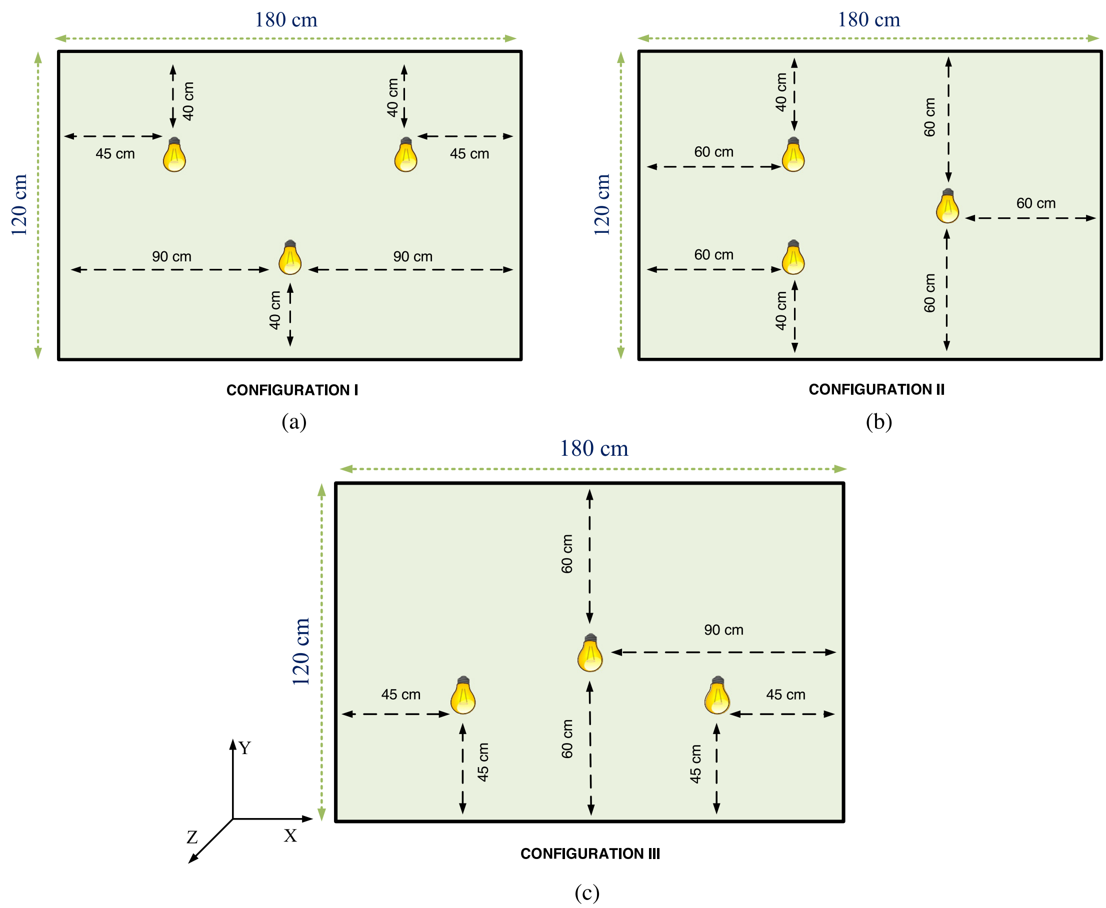


Fig. 2. Position of the Txs for the three configurations of: (a) I, (b) II and (c) III.

where $L = -\ln 2/\ln(\cos(\phi_{1/2}))$ is Lambertian order and $\phi_{1/2}$ is the Tx's semi-angle. ϕ and ψ are the radiant and incident angles, respectively. A and ψ_c denote the surface area and field of view (FOV) of the PD, respectively, d is the distance between the PD and the j^{th} LED and $T_s(\psi)$ is the optical filter gain.

2.3. Receiver

An optical Rx composed of a single PD and a trans-impedance amplifier (TIA) is used to regenerate the electrical m-CAP signal. A 5-CAP demodulator, which is exactly the inverse of the modulator (i.e., based on matched filters $g_I^n(t) = f_I^n(-t)$ and $g_Q^n(t) = f_Q^n(-t)$), is deployed to separate the sub-bands after removing the DC-bias. The first three sub-bands are then separated using an LPF for further processing to acquire the positioning data. The frequency content of these three sub-bands is extracted by Welch PSD [34] and compared with the values of the fingerprint map by Euclidean distance criterion, see Section 2.5 for the detailed generation process. The location corresponding to the least Euclidean distance is estimated as the position of the Rx (target) and the distance between the estimated and the actual positions of the Rx is measured to determine the PE. The 4^{th} and 5^{th} sub-bands are extracted by a band-pass filter (BPF) prior to demodulation and BER estimation.

2.4. Positioning algorithms

As mentioned in previous sections, various algorithms have been developed for VLP including trilateration, proximity, and fingerprinting. A brief description of these algorithms is explained here.

***(i)* Trilateration:** At least three LEDs with known positions are required for determining the position of the Rx. As shown in Fig. 3, the Rx first extracts the distance between each of the three Txs and the Rx (d_1, d_2, and d_3) by the RSS. Three circles with the centers of LED_1, LED_2 and LED_3 and the radii of d_1, d_2 and d_3, respectively are generated where the intersection of circles indicates the location of the Rx [4,11].

***(ii)* Proximity:** This algorithm is the simplest and the least accurate compared with others. In this method, the RSS values are calculated at the Rx and the highest values are chosen as the best approximation of the horizontal location of the Tx to the Rx. The accuracy of this algorithm can be improved by reducing the distance between the Txs or increasing the number of Txs [4,11].

***(iii)* Fingerprinting:** Offline and online steps are required for realization of the fingerprinting-based IPS, which is adopted in this paper. First, a room is divided into a self-defined grid containing several pixels namely the fingerprint locations. This is a laborious process; hence, choosing the appropriate step size (i.e., the grid size) is challenging. Based on AOA, RSS or TOA, the required parameter is extracted at the Rx and following processing is stored in a memory, which is referred to as the fingerprint map. Welch's method can be used to extract the PSD of the received signal. The measured parameter at the Rx is compared with the fingerprint map in order to determine the Rx's location. The Euclidean distance is usually used to determine the difference between the measured RSS values and those obtained from the fingerprint map [4,10].

2.5. Fingerprint map generation

First, the entire floor area of the room is converted into a grid corresponding to the step size. In order to investigate the impact of the step size, several values ranging from 1 to 20 cm are adopted in this work. The target is then placed at the first pixel's location, and the position indices (P_1, P_2 and P_3) are transmitted by the LEDs as

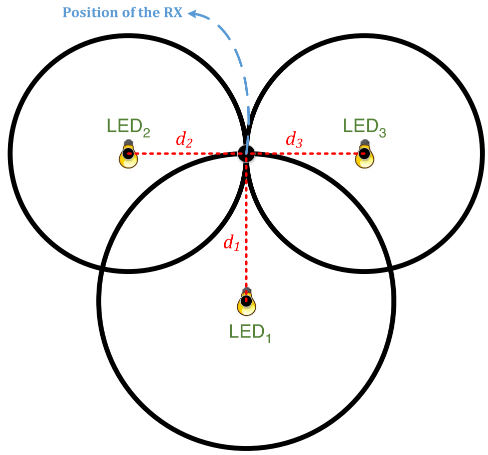


Fig. 3. Trilateration process.

described in Section 2.1. At the Rx, following separation of the sub-bands by the 5-CAP demodulator, the frequency spectrum of the first three sub-bands is extracted by Welch PSD for the target E_b/N_0 of 20 dB, as illustrated in Fig. 4 for the Rx positioned at the center of the floor. The areas under the PSD curves are calculated for each sub-band (i.e., 1–3), which corresponds to the positioning indices transmitted by the LEDs 1 to 3, respectively. The triplet generated is stored in the memory. Next, the Rx is placed at the next pixel and the same process is repeated until all pixels are evaluated. The fingerprint map is generated by putting together these triplets. Note, since the generated fingerprint map is different for the three configurations then the PE will also be different.

2.6. Pointing error

Fig. 5 illustrates the floor map of a room, where the number of squares is associated with the step size. The fingerprint map generated offline can be treated as a matrix, which is given by:

$$F = \begin{bmatrix} f_1^1 & f_2^1 & f_3^1 & \cdots & f_1^N & f_2^N & f_3^N \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\ f_1^{MN-N+1} & f_2^{MN-N+1} & f_3^{MN-N+1} & \cdots & f_1^{MN} & f_2^{MN} & f_3^{MN} \end{bmatrix}, \quad (7)$$

where $N = 180/\text{step size}$ and $M = 120/\text{step size}$. Assuming that, Welch PSD is a triplet of $\{f_1{}', f_2{}', f_3{}'\}$ for the received signal, the Euclidean distance between the triplet and the fingerprint map for all points on the grid is defined as:

$$d_{\text{euc}}^i = \sqrt{\left(f_1^i - f_1{}'\right)^2 + \left(f_2^i - f_2{}'\right)^2 + \left(f_3^i - f_3{}'\right)^2} \qquad for\ 1 \le i \le MN. \quad (8)$$

Next, we obtain $\min\left(d_{\text{euc}}^i\right) = d_{\text{euc}}^L$, where the index L refers to the position of the Rx. Thus, the PE is given by:

$$PE = \sqrt{\left(f_1^L - f_1\right)^2 + \left(f_2^L - f_2\right)^2 + \left(f_3^L - f_3\right)^2}, \quad (9)$$

where $\{f_1, f_2, f_3\}$ is the triplet of Welch PSD at the actual position of the Rx.

3. Results and discussion

In the previous sections, the prevalent techniques in IPSs as well as the proposed method of m-CAP VLP and the fingerprinting algorithm were introduced. In this section, we present computer simulation results for the hybrid VLP-VLC system, which are obtained using MATLAB and the real world parameters, see Table 1.

In the proposed system, the first three and last two sub-bands in 5-CAP are used for VLP and VLC, respectively. We have employed a

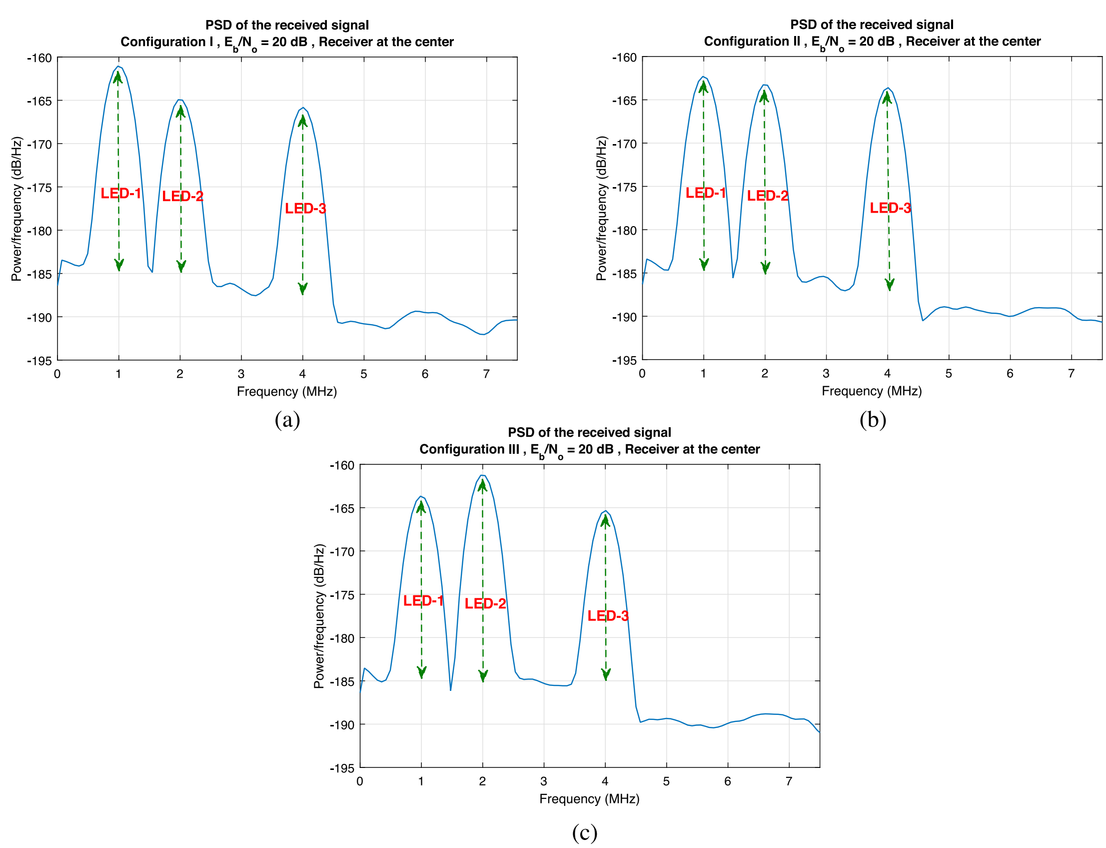


Fig. 4. Welch PSD at the Rx for three configurations: (a) I, (b) II and (c) III for generating the fingerprint map.

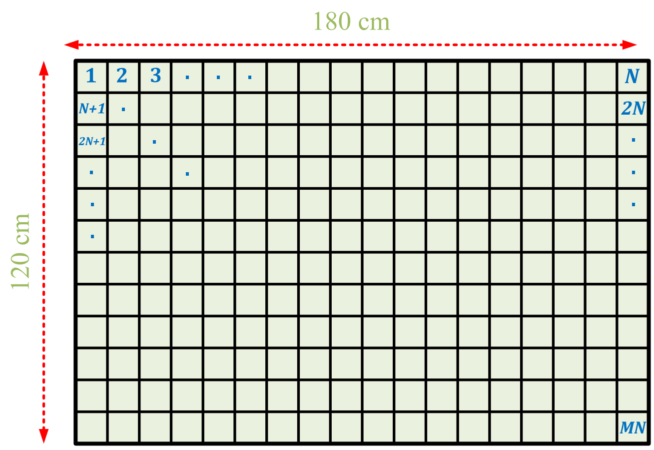


Fig. 5. A self-defined $M \times N$ pixels grid of the floor.

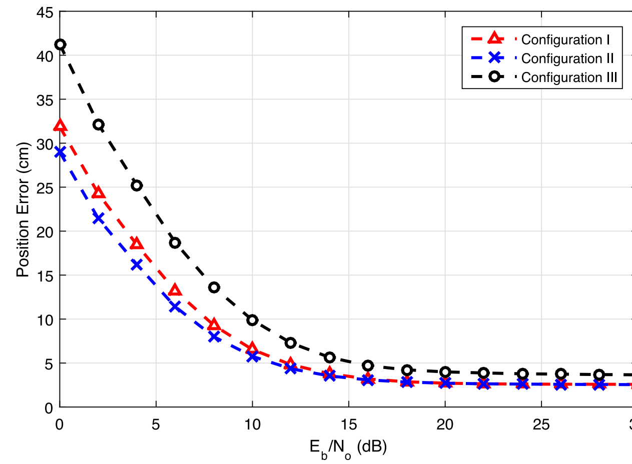


Fig. 7. PE vs. E_b/N_0 for different Tx's configurations.

fingerprint map generated offline to determine the Rx's position. This is accomplished by separating the first three sub-bands and determining their PSDs as shown in Fig. 4, in which the Rx is positioned at the middle of the floor. The same procedure is carried out for all points on the floor for the three configurations. For each point, the area under the PSD curve for each three sub-bands are calculated and stored as a triplet. Note, the number of triplets is equal to the number of pixels on the floor. The first three components (i.e., 1–3) of each triplet are associated with RSS at that point received from LED_1 to LED_3, respectively. By separating the components of all triplets, the normalized RSS values are obtained for all points on the floor, which corresponds to each LED as depicted in Fig. 6 for the configuration I, hence the equivalent fingerprint map is constructed. By comparing Figs. 6 and 2(a), we observe that, the RSS values for points close to the LEDs (i.e., just underneath the LEDs) are much higher than those at the corners of the room. Meanwhile, the accurate position of each LED can be easily identified by finding the maximum RSS values.

Figs. 7 and 8 illustrate the PE and relative PE, respectively as a function of E_b/N_0 for the three configurations. The relative PE is defined as the ratio of absolute PE (Eq. (9)) to the step size. As can be seen, both the PE and relative PE have improved with E_b/N_0. Below the E_b/N_0 of 10 dB, the configuration II offers the lowest PE followed by the I and III configurations. Similar to the absolute PE, configurations I and II outperform the configuration III by approximately 3 dB in terms of E_b/N_0 for $E_b/N_0 < 10$ dB.

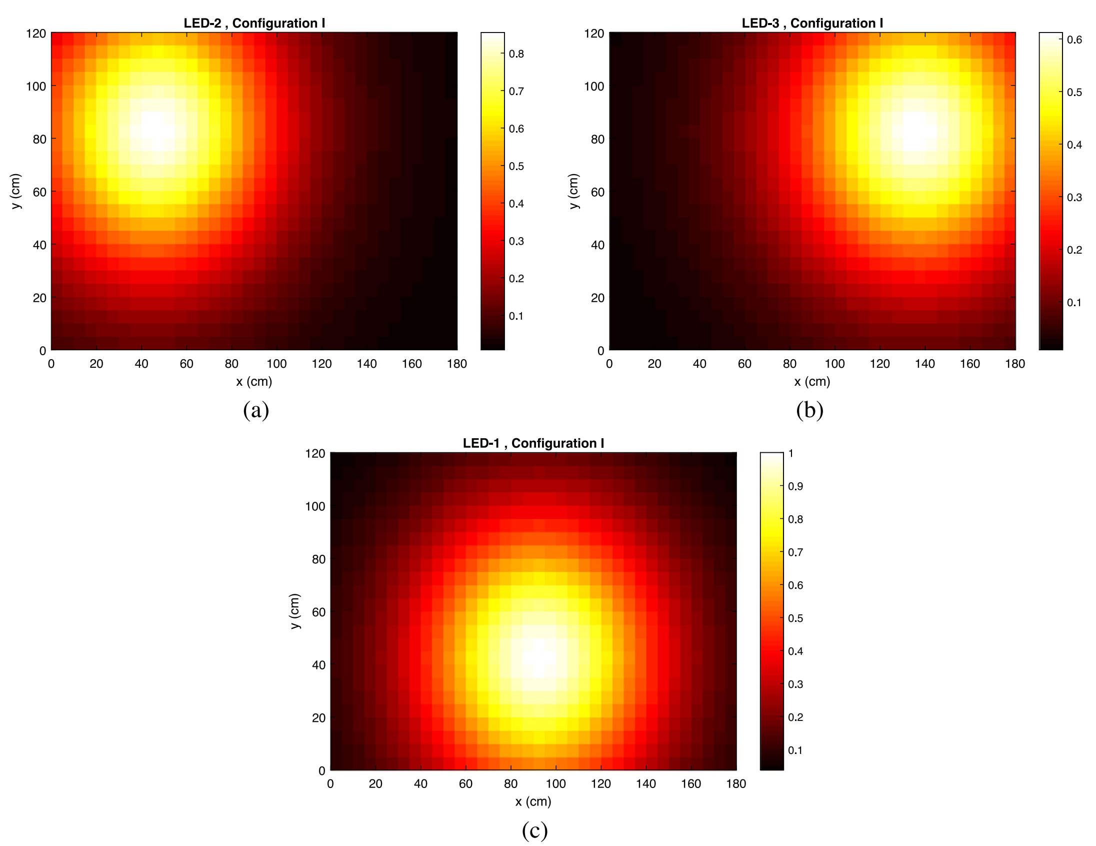


Fig. 6. The normalized RSS values for all points on the floor transmitted for the configuration I and for LEDs: (a) 2, (b) 3 and (c) 1.

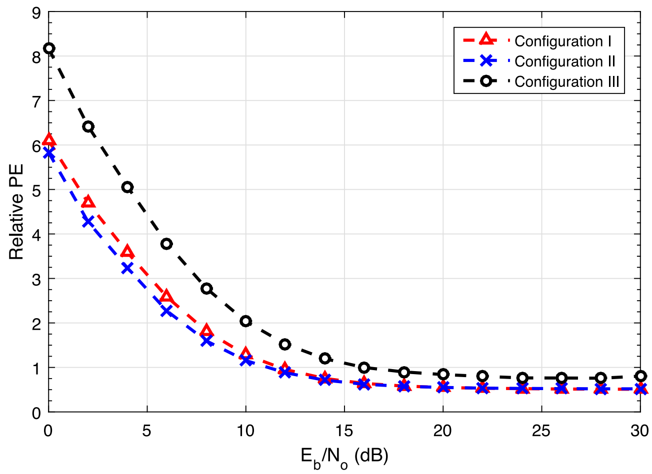


Fig. 8. The relative PE vs. E_b/N_0 for configurations I, II and III.

The main reason for the difference in the performance of these configurations is that the luminance of all points of the room is different for different configurations, see Fig. 9. As can be intuitively observed from this figure, the number of dimmer pixels of the configuration III are slightly more than those of configurations I and II. Also, most of the dark pixels in configurations I and II are in the lower half and right half of the floor, respectively since there is only one LED at those pixels, see Fig. 2. Hence, it can be concluded that configurations I and II, unlike the configuration III, provide adequate light for the entire room. Another important challenge in the fingerprinting-based positioning system is the fingerprint map values, which is perfectly symmetrical and there is more than one solution in calculating the position of the Rx provided the Txs are positioned on a straight line (horizontal or vertical) i.e., the configuration III. The reason for not using an exactly linear scenario is that, the PE will be so high therefore no need for the investigation. Note, for $E_b/N_0 > 15$ dB the PEs are almost the same for

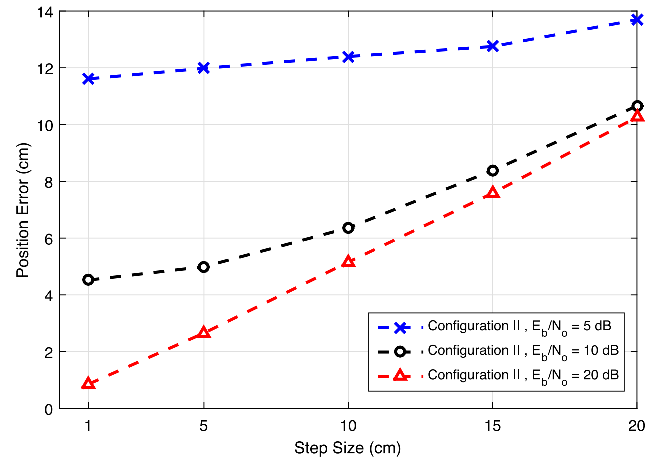


Fig. 10. PE vs. step size for different Tx's configurations at $E_b/N_0 = \{5 , 10 , 20\}$ dB.

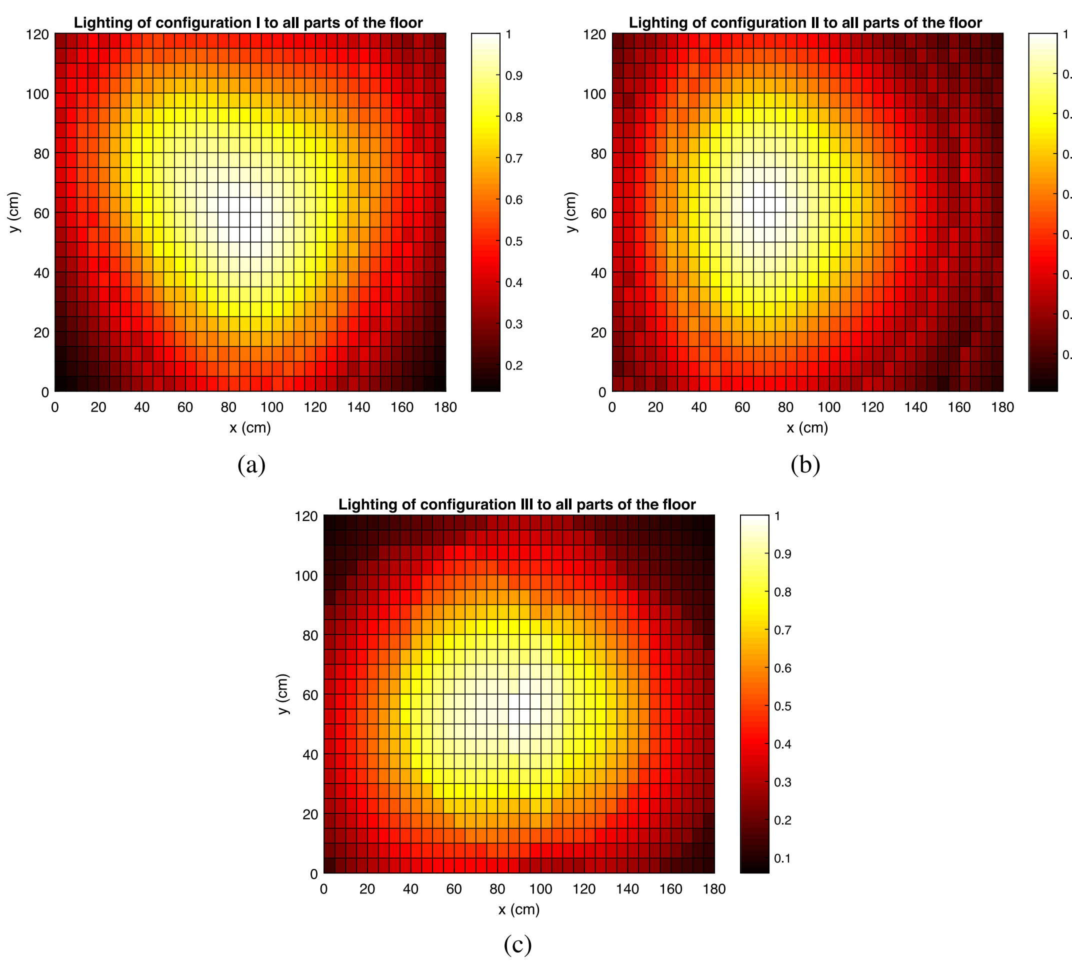


Fig. 9. A measure of illumination level to all points of the floor for configurations (a) I, (b) II and (c) III.

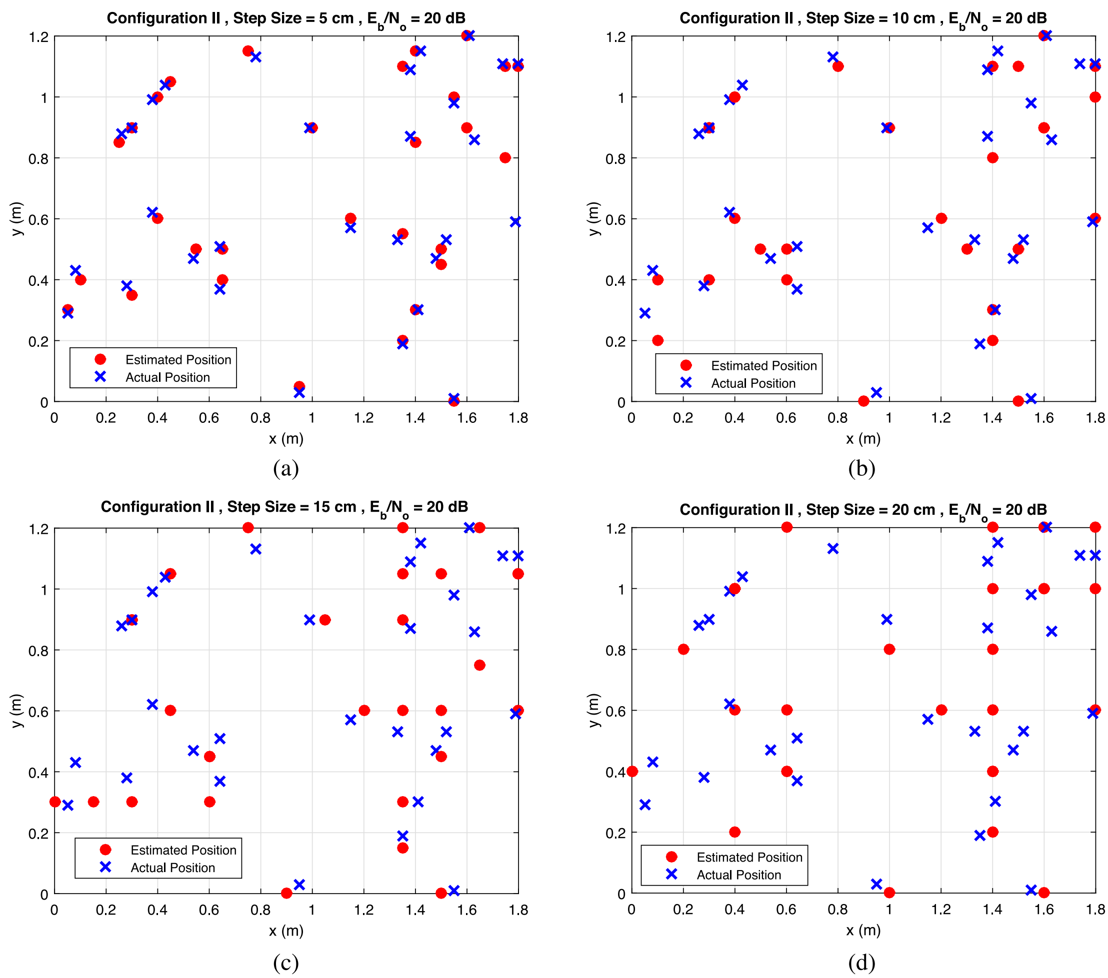


Fig. 11. Actual and estimated positions of the Rx for configuration II and for $E_b/N_0 = 20$ dB for the step sizes of: (a) 5, (b) 10, (c) 15 and (d) 20 cm.

all three configurations reaching the saturation level below 20 dB. For example, for E_b/N_0 of 20 dB, which is the default value adopted in this work, the PE value for configurations I and II is ~2.7 cm compared with 4 cm for the configuration III. Hereafter, the configuration II is used as the default configuration.

In fingerprinting-based positioning systems, selecting the step size is challenging task since it sets the limiting parameter in achieving the lowest PE level. Here, we have investigated five step sizes of {1, 5, 10, 15, 20} cm. Fig. 10 illustrates the average PE versus the step size for the default configuration II and for E_b/N_0 values of 5, 10 and 20 dB. Note, the PE increases with the step size. This is because that increasing the step size (i.e., the bigger pixels) means a grid with a reduced number of pixels, which will result in higher quantization errors (QE). For a given step size, PE improves with E_b/N_0. As an example, for the step sizes of 1 and 20 cm and E_b/N_0 of 5 and 20 dB, the drop in PE values are ~11 and 3.5 cm, respectively. In fact, with increasing the step size, the QE overcomes the noise-induced error. Note, as can be seen in Table 2, lower PE using smaller step sizes can be achieved, but at the cost of increased computational complexity. Note, the default value for the step size adopted here is 5 cm. Fig. 11 shows the actual and estimated positions for the configuration II, E_b/N_0 of 20 dB and for a range of step sizes. It can be intuitively observed that, the difference between the actual and the estimated positions increases with the step size. Fig. 12 demonstrates this difference as a quantitative point of view rather than intuitively observation. The PE values for each point within the room are different due to the uneven luminance levels as shown in Fig. 13. Note that, the Tx configuration, the step size value and E_b/N_0 are assumed to be the default values. As can be seen, the maximum PE of 8.4 cm are observed at the corners of the room whereas the minimum PE of 1 cm is at a location just below LED_2. Also, the average PE for the configuration II with a step size of 5 cm and E_b/N_0 of 20 dB is approximately 2.65 cm.

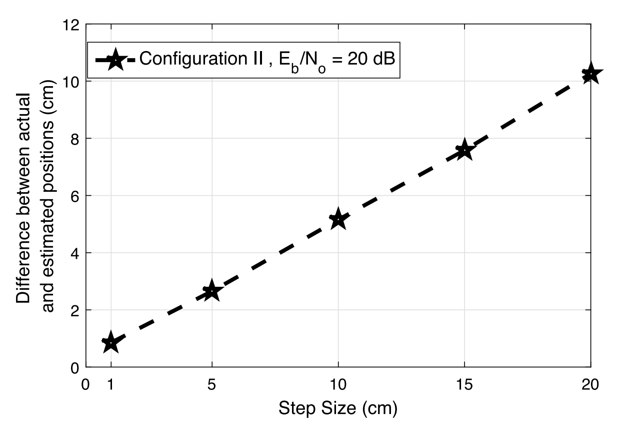


Fig. 12. The average difference between actual and estimated positions vs. step size values for configuration II and $E_b/N_0 = 20$ dB.

Finally, the BER performance of the VLC link is evaluated for different Tx configurations as depicted in Fig. 14. As can be observed, configurations I and II demonstrate almost the same BER performance whereas the configuration III offers the worst performance because

Table 1
System parameters.

Symbol	Parameter	Value
	Room size	180 cm × 120 cm × 100 cm
$\varphi_{1/2}$	Semi-power half angle	70°
FOV	PD field of view	85°
A	PD detector area	1 cm^2
BW_{PD}	3-dB bandwidth of PD	10 GHz
R	Responsivity of PD	0.54 A/W
$T_s(\psi)$	Optical filter gain	1
m	Number of sub-bands	5
f_{LED}	LED 3-dB cut-off frequency	4.5 MHz
β	Roll-off factor of I/Q filters	0.5
n_s	Up-sampling factor	15
L_f	I/Q filters length	10 Symbols
BW_{tot}	Total bandwidth	7.5 MHz
BW_{sub}	Sub-band BW	1.5 MHz
D_4 & D_5	Modulation type	PSK
M	Modulation order	2
P_1	Positioning index of LED$_1$	$-1-i$
P_2	Positioning index of LED$_2$	$1+i$
P_3	Positioning index of LED$_3$	$1-i$
	Step size	1, 5, 10, 15, and 20 cm
	LEDs position of configuration I	$\begin{cases} \mathrm{LED}_1: (90,40,100) \\ \mathrm{LED}_2: (45,80,100) \\ \mathrm{LED}_3: (135,80,100) \end{cases}$
	LEDs position of configuration II	$\begin{cases} \mathrm{LED}_1: (60,40,100) \\ \mathrm{LED}_2: (60,80,100) \\ \mathrm{LED}_3: (120,60,100) \end{cases}$
	LEDs position of configuration III	$\begin{cases} \mathrm{LED}_1: (45,45,100) \\ \mathrm{LED}_2: (90,60,100) \\ \mathrm{LED}_3: (135,45,100) \end{cases}$
BW_{TIA}	Bandwidth of TIA	10 MHz
G_{TIA}	TIA gain	60 dB
	LPF cut-off frequency	4.5 MHz
	BPF pass-band range	4.5 – 7.5 MHz
	LPF and BPF type and order	5th order Butterworth filter

Table 2
Comparison between complexity and PE.

Step size (cm)	Grid pixels (#)	Simulation time to find target position (s)	PE at $E_b/N_0 = 20$ dB (cm)
1	21 901	0.1225	0.85
5	925	0.0051	2.65
10	247	0.0014	5.15
15	117	0.0007	7.58
20	70	0.0004	10.26

of insufficient illumination on floor surface. For the forward error correction BER of 3.8×10^{-3}, the gain in E_b/N_0 for the configurations I and II is ∼4 dB compared with the configuration III.

4. Conclusion

In this paper, the *m*-CAP based indoor hybrid VLP and VLC system was proposed and evaluated by using the fingerprinting algorithm for the first time. The results were compared for different Tx's configurations and a range of step sizes in terms of the PE and BER performances. We showed that, the PE performance of the system is improved by increasing E_b/N_0. The comparison between configurations showed that, configurations I and II outperformed the configuration III since it is approximately linear and therefore the resultant fingerprint map is more symmetrical. For example, at E_b/N_0 of 5 dB the PE values are 14, 16 and 22 cm for configurations II, I and III, respectively. We also showed that, PE increases with the step size, which is the most critical parameter in fingerprinting-based positioning systems, for a fixed E_b/N_0. For a fixed E_b/N_0 of 10 dB the PE values are 4.5 and 10.7 cm for step sizes of 1 and 20 cm, respectively. In contrast, for

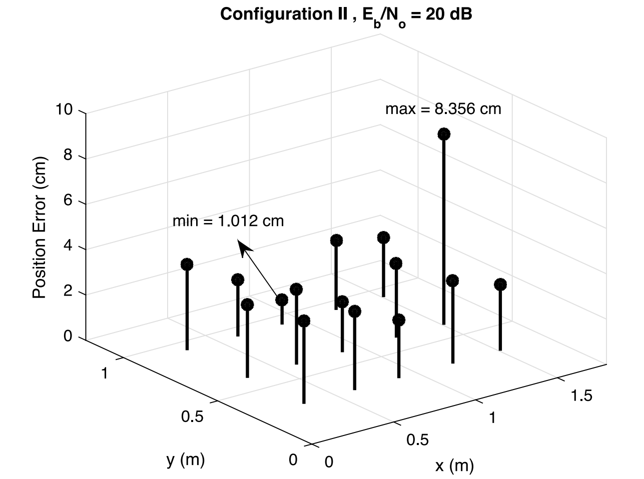


Fig. 13. PE values at different points for the configuration II and for the E_b/N_0 of 20 dB.

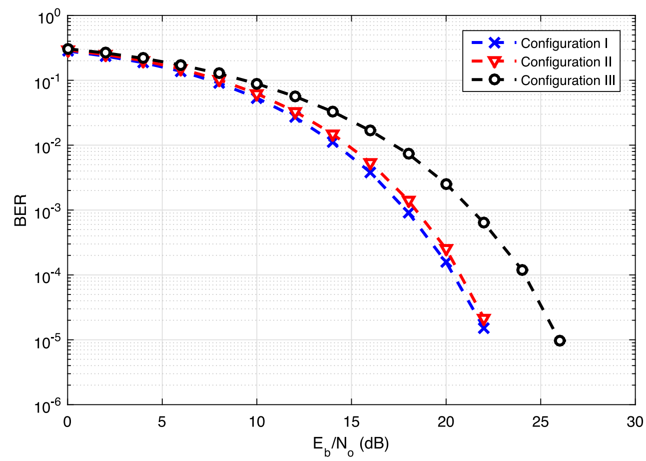


Fig. 14. BER vs. E_b/N_0 for VLC and for different Tx's configurations.

a fixed step size the PE decreased with increasing E_b/N_0. Numerical results showed that, the PE for a fixed step size of 5 cm were 13.8 and 3.3 cm for E_b/N_0 of 5 and 15 dB, respectively. It was also demonstrated that, the PE values were different for different locations on the floor of the room due to the lower and higher luminance levels at the corners and center of the room, respectively. Therefore, we adopted the average PE for all points to assess the system performance. Finally, we investigated the BER performance of the VLC link for different Tx configurations, and showed that configurations I and II offered lower BER values of 1.56×10^{-4} and 2.52×10^{-4}, respectively compared with the configuration III (i.e., 2.54×10^{-3}) because of higher illumination levels.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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